TOWARDS EVIDENCE-AWARE & TRANSPARENT ECG AI

Synthesis of Completed Foundations, Evidence Observability Dilemma, and Clinical Verification Roadmap

1. ESTABLISHED FOUNDATIONS

End-to-End Control Pipeline (Modules 1 - 4)

MODULE 01 — 12-Lead Ingestion

Foundational Data Pipeline

Standardized multi-cohort ingestion (PTB-XL, Chapman, MIMIC-IV), bandpass filtering, and 12-lead tensor curation.

MODULE 02 — Physiological Masking

Lead-Degradation Framework

Controlled physiological sensor masking simulating real-world reduced-lead telemetry without artificial waveform inpainting.

MODULE 03 — Z_ECG Latents

ECG-FM Representation Pipeline

Frozen 90.8M parameter foundation model extracting 768-d Z_ECG latents paired with rule-extracted clinical observations.

MODULE 04 — Control Reasoner

ECG-Only Control Model

Baseline clinical reasoning transformer evaluating cross-attention traces and diagnostic predictions from ECG signals alone.

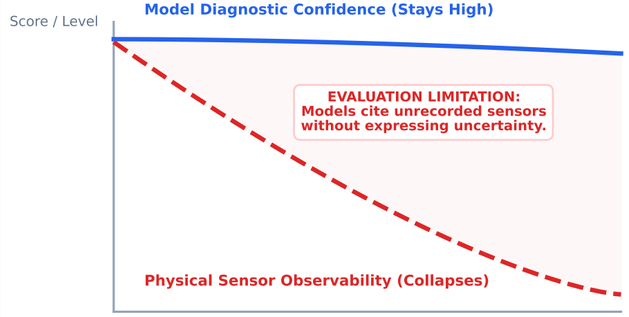
STATUS: Validated Baseline Architecture Complete

2. THE CORE RESEARCH PROBLEM

Evidence Sufficiency vs. Epistemic Uncertainty

THE SENSOR-EVIDENCE INTEGRITY GAP

Observed Behavior Under Lead Degradation (12-Lead to Reduced Leads)



DESIRED CLINICAL DECISION BEHAVIOR

Guiding Models Between Confident Claims and Honest Abstention

REGIME A — **Sufficient Evidence -> Support Prediction**

When required anatomical leads are active & verified:
System produces a confident, well-grounded diagnostic prediction.

REGIME B — **Evidence Degraded -> Express Uncertainty**

When critical leads are missing & vectors unobservable:
System abstains from hallucinating & signals epistemic uncertainty.

CORE MISSION: Distinguish Grounded Insight from Overconfident Guesswork

3. NEXT STEPS & FUTURE ROADMAP

Towards Transparent, Evidence-Aware AI

NEXT STEP 01 — In Development

Clinical Knowledge-Assisted Path

Incorporate structured electrophysiological priors (SNOMED-CT relations) to reroute attention toward observable alternate leads during degradation.

Architecture: SNOMED-CT Priors + Frontal Re-routing

NEXT STEP 02 — Gate Design Ready

Evidence-Grounding Checks

Deploy deterministic ternary verification gate classifying claims as SUPPORTED, CONTRADICTED, or UNAVAILABLE against active hardware masks.

Ternary Gate: [SUPPORTED] [CONTRADICTED] [UNAVAILABLE]

NEXT STEP 03 — Protocol Specified

Controlled Protocol Comparison

Execute multi-cohort comparative benchmarking under formal clinical protocols to quantify safety boundaries, RFB thresholds, and trade-offs.

Audit Protocol: Reasoning Failure Boundary (RFB) Benchmark

ULTIMATE GOAL: Transparent & Evidence-Aware Medical AI

Figure: Conceptual Framework Synthesizing Established Foundations, Evidence Observability Gap, and Transparent Evaluation Roadmap | Batch 43 (SSN CSE)